Chapter 2: Sound and Audio

Basic Sound Concepts

- Sound is a physical phenomenon produced by the vibration of matter, such as a violin string, or a block of wood. This vibration creates pressure fluctuations in the surrounding air, producing pressure waves that propagate outward.
- The oscillating pattern of these waves, known as a waveform, as illustrated in figure below. We perceive sound when these pressure waves reach our ears.
- The waveform repeats the same shape at regular interval and this portion is called period. Since, sound wave is never perfectly smooth or uniformly periodic.

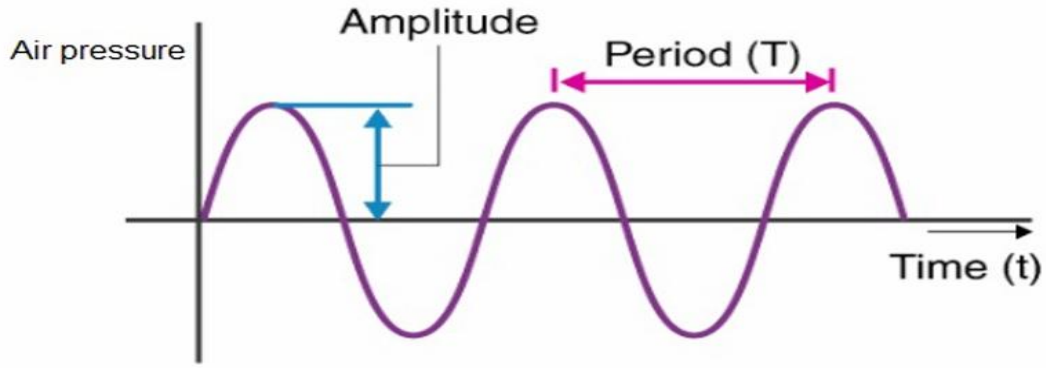


Figure: Sound waveform

- When a sound has a noticeable periodicity, it is often classified as music, whereas **sound without regular patterns** is simply referred to as noise.
- Examples of periodic sounds sources are musical instruments, vowel sounds, the whistling wind and bird songs. On the other hand, non-periodic sounds include drum beats, coughing, sneezing or the sounds of water flowing.
- Fundamental properties/ characteristics of sound are frequency and amplitude.

1. Frequency

- The frequency of a sound is the reciprocal value of the period, representing the number of cycles per second and measured in hertz (Hz) or kilohertz (kHz).
- The frequency range of sound is divided into

Infra-sound	0-20 Hz
Human hearing frequency range	20 Hz – 20kHz
Ultrasound	20kHz – 1GHz
Hypersound	1GHz – 10THz

- The audiosonic range (20Hz – 20kHz) is essential for multimedia, encompassing signals like speech and music. Audio in this range is also called acoustic signals.
- Noise, defined as sound without specific function, can be added to speech and music, though the term is flexible and can include unintelligible language as well.

- Wavelength is the distance travelled in one cycle. The wavelength (λ) is given by:

$$\lambda = \frac{v}{f}$$

Where, v: velocity and f: frequency

2. **Amplitude**
   - A sound has a property called amplitude, which humans perceive subjectively as loudness or volume.
   - The amplitude of a sound is a measuring unit used to deviate the pressure wave from its mean value (idle state).

Sound Representation

1. **Sampling**

- The smooth, continuous curve of a sound waveform is not directly represented in a computer. To represent the continuous curve of a sound wave on a computer, the computer must first measure the amplitude of the waveform at regular time intervals.
- It then uses these measurements to create a sequence of sampling values, commonly called samples. The period of a digitally sampled wave is shown in figure below.
- The component that converts an audio signal into a sequence of digital samples is known as an ADC (Analogy-to-Digital Converter). The reverse conversion is performed by a Digital-to-Analogy Converter (DAC).

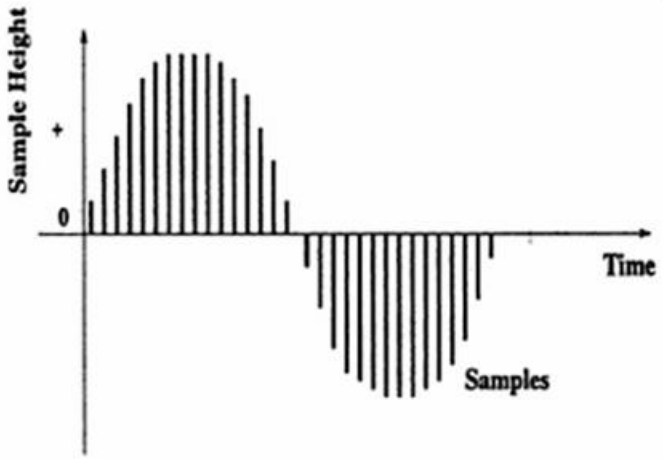


Figure: - Sample waveform

Sampling Rate

- The sampling rate is the frequency at which a continuous waveform is sampled, measured in Hertz (Hz). The CD sampling rate of 44,100 Hz means that the waveform is sampled 44100 times per second, which slightly exceeds the human hearing range.

- Due to the Nyquist theorem, the maximum usable frequency is half of the sampling rate. Therefore, a CD's sampling rate of 44,100 Hz converts frequencies up to 22,050 Hz, which is just within the human hearing limit.

2. Quantization

- It is the process of mapping a large set of input values to smaller set. The quantization of the sample value depends on the number of bits used in measuring the height of the waveform.
- Digitizing an analog signal involves two steps: sampling and quantization. Sampling captures discrete values at regular intervals while quantization converts these samples into a limited number of levels.
- For instance, 8-bit quantization provides 256 levels, whereas 16-bit (CD-quality) provides over 65,536 levels. Figure below illustrates a 3-bit quantization example.

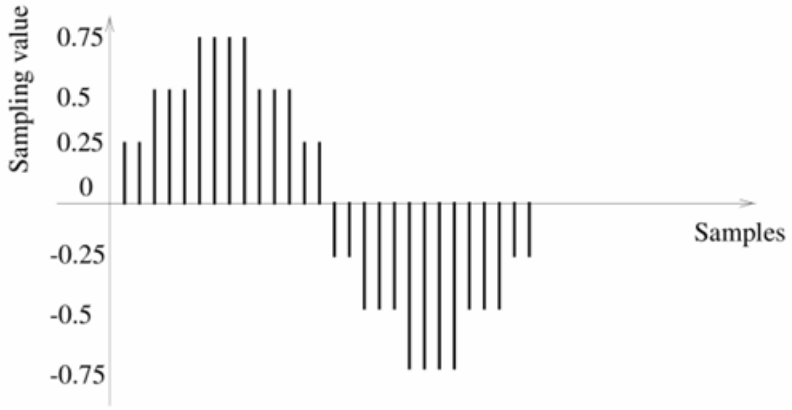


Figure: 3-bit quantization

With 3-bit quantization, values are limited to eight distinct levels: 0.75, 0.5, 0.25, 0, -0.25, -0.5, -0.75, and -1. Lower bit depth results in a more angular waveform and generally reduced sound quality, as fewer bits represent the original signal less precisely.

Formats

- Sound formats define how digital audio data is encoded, stored, and compressed. Each format has unique characteristics that make it suitable for different applications, such as streaming, archiving, or high-quality playback.

1. Uncompressed Formats:

- Uncompressed audio formats preserve the original audio quality without reducing file size, making them large but highly accurate. These formats are ideal for professional audio production or high-quality audio archiving.

  WAV (Waveform Audio File Format): A widely-used uncompressed format developed by Microsoft and IBM, WAV files store audio in raw, uncompressed PCM (Pulse Code Modulation) form. It is a standard format for CDs and is often used in professional audio editing due to its high-quality and broad compatibility.

AIFF (Audio Interchange File Format): Created by Apple, AIFF is similar to WAV in terms of quality and file size. It is often used on Apple devices and by audio professionals requiring lossless quality.

2. **Lossless Compressed Formats**

- Lossless compression reduces the file size while preserving the original audio quality. These formats are beneficial for listeners and professionals who want high-quality audio without the large file size of uncompressed formats.

  FLAC (Free Lossless Audio Codec): A popular open-source format that compresses audio without any loss in quality. FLAC files are widely supported, and the format is preferred for high-resolution audio storage and streaming.

  ALAC (Apple Lossless Audio Codec): Apple's version of a lossless codec, ALAC compresses audio while maintaining quality. It is compatible with iTunes and other Apple devices, making it a common choice for Apple users.

3. **Lossy Compressed Formats**

- Lossy compression significantly reduces file size by discarding some audio data, typically focusing on parts less perceivable to human hearing. While this reduces quality slightly, it makes these formats ideal for online streaming and limited storage.

  MP3 (MPEG Layer-3 Audio): Perhaps the most well-known audio format, MP3 offers high compression rates, which makes it ideal for streaming and storage on devices with limited space. While MP3 does sacrifice some quality, it remains popular for its efficiency and compatibility.

  AAC (Advanced Audio Codec): A successor to MP3, AAC generally offers better sound quality at similar bit rates. It is widely used in online streaming services like YouTube and Apple Music, as well as on most smartphones and computers.

Basic Music Concepts

- Music is a form of sound that is artistically arranged using elements like rhythm, melody, and harmony to create a pleasing or meaningful auditory experience.
- The relationship between music and computers has become more and more important especially considering the development of MIDI (Music Instrument Digital Interface).
- The MIDI interface between electronic musical instruments and computers is a small piece of equipment that plugs directly into the computer's serial port and allow the transmission of music signals.

Introduction to MIDI

- MIDI is a set of specifications used in building their instruments to enable easy musical information exchange between instruments form different manufacturers.

- The MIDI protocol works as a complete music description language in binary form, with each word describing a musical performance action assigned a specific binary code.
- A MIDI interface has two different components:

1. **Hardware to connect equipment:** MIDI hardware specifies the physical connection of musical instruments, adding a MIDI port to an instrument, defining a MIDI cable for connecting two instruments, and processing the electrical signals received over the cable.

2. **Data format for processing information:** The MIDI data format encodes information processed by the hardware, but it does not include individual sampling values as seen in audio data formats. Instead, it encodes specific details for each instrument, such as the start and end of scores, base frequency, loudness, and the instrument used.

The MIDI data format is digital, with data grouped into MIDI messages. When a musician presses a key, the MIDI interface generates a MIDI message that indicates the start of a score and its intensity, transmitting this message to connected devices. When the key is released, another MIDI message is created and transmitted.

MIDI Devices:

- An instrument that follows both components of the MIDI standard is called a MIDI device (e.g., a synthesizer) and can communicate with other MIDI devices over designated channels. The MIDI standard specifies 16 channels, with each MIDI device (musical instrument) mapped to a specific channel.
- Musical data sent over a channel is reproduced by the synthesizer on the receiver's end. The MIDI standard also defines 128 instruments using numbers, including various noise effects (e.g. a phone ringing or an airplane taking off). For example, 0 represents a piano, 12 a marimba, 40 a violin, and 73 a flute.
- The core of any MIDI system is the MIDI synthesizer device. Most synthesizers have the following common components:
    1. **Sound Generators:** Sound generators perform the actual task of synthesizing sound. The rest of the synthesizer's components are primarily used to control the sound generator.
    2. **Microprocessor:** The microprocessor communicates with the keyboard to know what notes the musician is playing and with the control panel to interpret the musician's commands. It also sends and receives MIDI messages.
    3. **Keyboard:** The keyboard provides the musician with direct control over the synthesizer. Pressing keys on the keyboard signals the microprocessor what notes to play and how long to play them.
    4. **Control Panel:** The control panel manages functions that are not directly related to notes and durations (which are controlled by the keyboard). Panel control includes: a slider that sets overall volume of the synthesizer, a button that turns the synthesizer on and off, and a menu that calls up different patches for the sound generators to play.

5. **Auxiliary Controllers:** Auxiliary controllers offer additional control over the notes played on the keyboard. Two common controls on a synthesizer are pitch bend and modulation.
6. **Memory:** The synthesizer's memory stores patches for the sound generators and settings for the control panel.

MIDI Messages

- MIDI messages transmit information between MIDI devices and determine what kinds of musical events can be passed form device to device. The format of MIDI messages consists of the status byte (the first byte of any MIDI message), which describes the kind of message, data bytes (the following bytes).
- MIDI messages are divided into two different types: **Channel Messages** and **System Messages.**
  1. **Channel Messages:** Channel messages go only to specified devices. There are two types of channel messages: **Channel Voice Messages** and **Channel Mode Messages.**
     - **Channel Voice Messages** send actual performance data between MIDI devices, describing keyboard action, controller action, and control panel changes. They describe music by defining pitch, amplitude, timbre, duration, and other sound qualities. Each message has at least one and usually two data bytes that accompany the status byte to describe these sound qualities. Example: Note On, Note Off, Channel Pressure, Control change, etc.
     - **Channel Mode Messages** determine the way that a receiving MIDI device responds to channel voice messages. They set the MIDI channel receiving modes for different MIDI devices, stop spurious notes from playing, and affect other aspects of MIDI control. Examples are All Notes Off, Omni Mode Off, etc.
  2. **System Messages:** System messages go to all devices in a MIDI system because no channel numbers are specified. There are three types of system messages: **System Real-Time Messages, System Common Messages,** and **System Exclusive Messages.**
     - **System Real-Time Messages** are vital, short and simple, consisting of only one byte. They carry extra data with them. These messages synchronize the timing of MIDI devices in performance; therefore, it is important that they be sent at precisely the time they are required. To avoid delays, these messages are sent in the middle of other messages if necessary. Examples of such messages are System Reset, Timing Clock (MIDI clock), etc.
     - **System Common Messages** prepare devices to start or synchronize a song, such as selecting a song or tuning synthesizers. Examples include Song Select and Tune Request.
     - **System Exclusive Messages** allow MIDI manufacturers to create customized MIDI messages to send between their MIDI devices. This coding starts with a system-exclusive-message, where the manufacturer is specified, and ends with end-of-exclusive messages.

MIDI Standards:

The MIDI standard is a technical protocol that allows electronic musical instruments, computers, and other equipment to communicate, control, and synchronize with each other. Developed in the early 1980s by a coalition of major instrument manufacturer, MIDI transformed music production by enabling the integration and control of diverse musical instruments and devices from different manufacturers.

1. **MIDI Message Structure:** MIDI messages are small packets of data that are transmitted in real-time. They contain instructions about which notes to play, how long to play them, and various other parameters.
2. **MIDI Channels:** The MIDI standard provides 16 channels per MIDI connection. This allows multiple instruments or sounds to be controlled independently over a single MIDI cable. Each instrument or sound assigned to a specific channel (e.g. drums on channel 10, piano on channel 1). This setup allows a single MIDI stream to control multiple instruments simultaneously.
3. **MIDI Connectors and Hardware:** The original MIDI standard specifies the use of 5-pin DIN connectors and MIDI cables for connecting devices. There are three types of MIDI ports:
   MIDI In: Receives MIDI messages.
   MIDI Out: Sends MIDI messages generated by the instrument.
   MIDI Thru: Passes incoming MIDI messages to another device.
4. **General MIDI (GM) Standard:** This standard was introduced to ensure compatibility between MIDI files on different devices. It specifies a standardized set of 128 instruments and sounds (e.g., pianos, strings, drums) and assigns them to specific program numbers.
5. **MIDI Clock and Sync:** This includes a system for synchronizing devices, knows as MIDI Clock. This ensures that devices can stay in sync when playing back or recording.
6. **MIDI File Format:** The MIDI standard file format is Standard MIDI File (SMF), which allows MIDI data to be saved, edited, and replayed. SMF files are platform-independent and often use the mid file extension.
7. **MIDI Polyphonic Expression (MPE):** MPE is a standard that extends MIDI's expressiveness by allowing each note played on a device to have its own channel of control data.

MIDI Software

Once a computer is connected to a MIDI system, a wide variety of MIDI applications can be run on it. The software applications generally fall into four major categories:

1. **Music Recording and Performance Applications:** These applications allow the recording of MIDI messages as they are received from other MIDI devices, and also provide options for editing and playback of messages during performances.
2. **Musical Notation and Printing Applications:** These applications enable users to write music using traditional musical notation. The music can then be played back through a performance program or printed on paper for live performance or publication.

3. **Synthesizer Patch Editors and Librarians:** These programs allow storage of various synthesizer patches in the computer's memory and disk drives, and also facilitate patch editing within the computer.

The main issue in current MIDI-based computer music systems is interactivity. The processing chain of interactive computer music systems can be conceptualized in three stages:

1. **The Sensing Stage:** In this stage, data is collected from controllers that read gesture information from human performers on stage. (e.g., detecting finger pressure on keyboard)
2. **The Processing Stage:** Here, the computer reads and interprets the information from the sensors and prepares the data for the response stage. (e.g., convert pressure into MIDI message)
3. **The Response Stage:** In this final stage, the computer and various sound-producing devices work together to create a musical output.

Speech

Speech is the vocalized form of human communications. It can be perceived, understood, and generated by human and also by machines. A human adjusts himself/ herself very efficiently to different speakers and their speech habits.

Speech signals have two properties which can be used in speech processing:

a. Voice speech signals show during certain time intervals almost periodic behavior, making them quasi-stationary for about 30 milliseconds.
b. The spectrum of audio signals shows characteristic maxima, which are mostly 3-5 frequency bands. These maxima, called formats, occur because of resonances of the vocal tract.

Speech Generation/Synthesis

- Speech generation, also known as speech synthesis, is the process of creating artificial human speech that can be heard through a device or computer. In other words, speech generation is a process that converts text into an acoustic signal, making written language audible.
- An important requirement for speech generation is real-time signal generation. If such requirement met, a speech output system could transform text into speech automatically without any lengthy preprocessing.
- Some applications only need a limited vocabulary; an example is the spoken time announcement of a telephone answering service. However, most applications need a large vocabulary, if not an unlimited vocabulary.
- Generated speech must be understandable and must sound natural. The requirement of understandable speech is a fundamental assumption, and the natural sound of speech increases user acceptance.
- A computer system used for speech generation is called a speech synthesizer. Figure below shows the components of such a system.

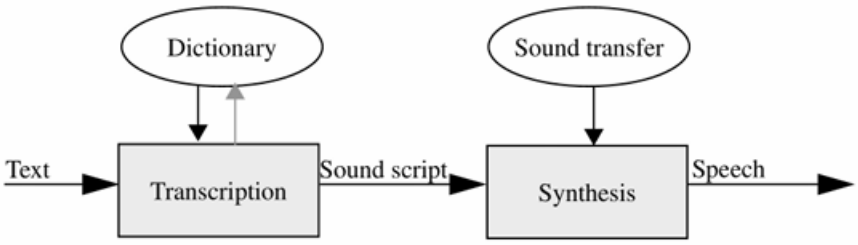


Figure: Components of a speech synthesis system, using sound concatenation in the time range

1. **Phonetic Transcription/ Generation of a Sound Script:** The first step involves a transcription, or translation, of the text into the corresponding phonetic transcription. Most methods use a lexicon containing a large quantity of words, syllables, or tone groups. The creation of such a library is extremely complex and can be an individual implementation or a common lexicon used by several people.
2. **Acoustic Signal Conversion/Generation of speech:** The second step converts the phonetic transcription into an acoustic speech signal, where concatenation can occur in the time or frequency range. While the first step normally has a software solution, the second step often requires signal processors or dedicated hardware processors.

Reproduced Speech Output

There are two way of speech generation/output performed by time-dependent sound concatenation and a frequency-dependent sound concatenation. [**see page 44, 45 and 46 of book (Pearson-Ralf Steinmetz and Klara Nahrstedt).**

Speech Analysis:

- Speech analysis/ input deals with the research areas as show in figure below:

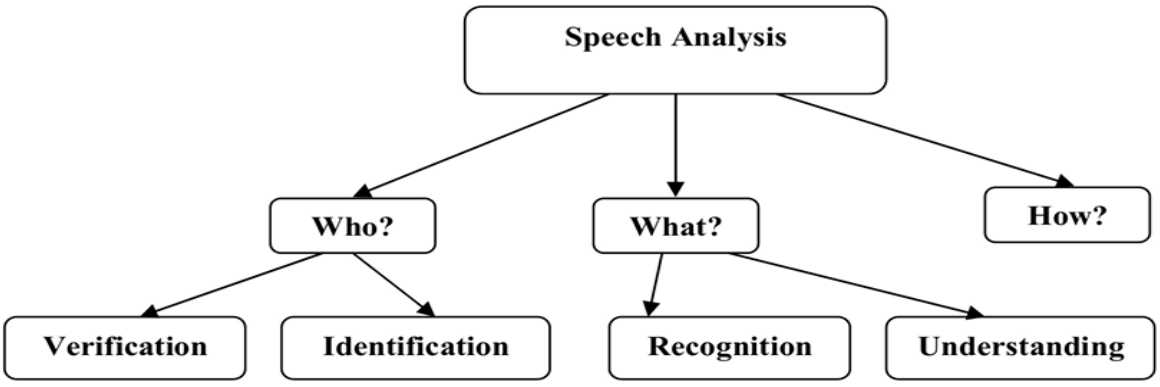


Figure: Research areas of speech analysis

- Human speech has certain characteristics determined by a speaker. Hence, speech analysis can serve to analyze who is speaking, i.e., to recognize a speaker for his/her identification and verification. The computer identifies and verifies the speaker using an acoustic fingerprint. An acoustic fingerprint is a digitally stored speech sample (e.g. a specific phrase) from an individual.
- Another main task of speech analysis is to analyze what has been said, i.e., to recognize and understand the speech signal itself. Based on sequence of speech the corresponding text is generated. This can lead to a speech-controlled typewriters, translation tools, or assistive devices for individuals with disabilities.
- Another area of speech analysis tries to research speech patterns with respect to how a certain statement was said. For example, the tone of a spoken sentence can vary depending on whether the speaker is calm or angry.
- The primary goal of speech analysis correctly determines individual words with probability less than or equal to 1. A word is only recognized with a certain probability. Therefore, environmental noise, room acoustics and speaker's physical and psychological conditions play important role.

Speech Recognition

- Speech recognition is the process of transforming an acoustic signal, picked up by a microphone or telephone, into a sequence of words or a similar format. Speech recognition systems are classified into speaker-independent and speaker-dependent systems. Speaker-dependent system, which are pre-trained, can recognize with higher accuracy, while speaker-independent systems can recognize with lower accuracy.

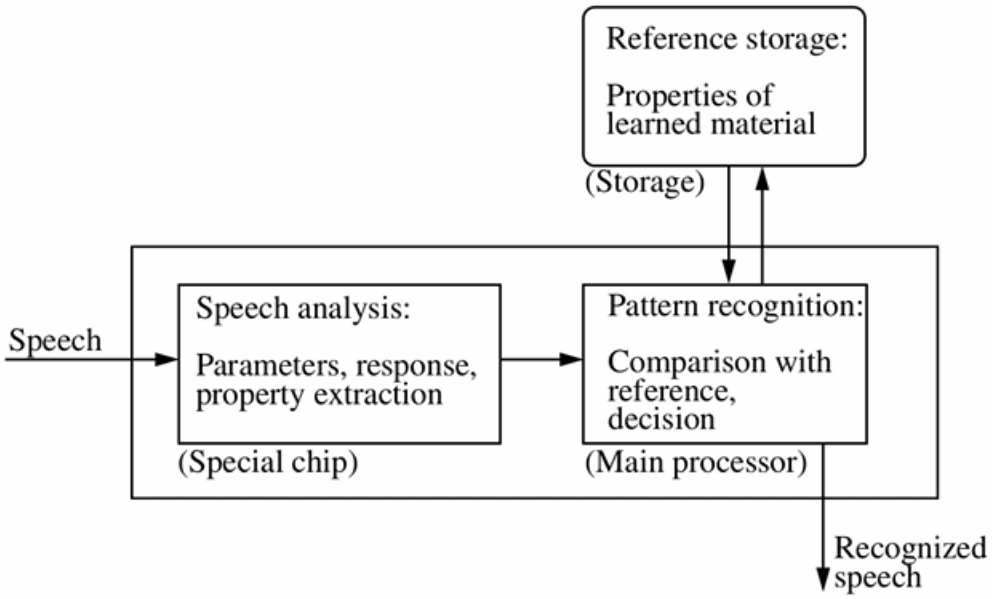


Figure: The speech recognition principle: the tasks are distributed to system components by the basic principle "extract characteristics to reduce data".

- The principle of speech recognition is shown in figure above which involves comparing distinctive features of individual utterances with a set of previously extracted speech elements.
- Typically, these features are quantized, allowing for a structured analysis of the speech sequence. The result is then matched against known references to classify it within one of the existing speech elements. Once identified, utterances can be stored, transmitted, or processed as a parameterized sequence of speech components.
- The principle illustrated in above figure can be applied multiple times, each iteration focusing on different characteristics.
- The application of this principle, as depicted in above figure can be broken down into the steps shown in figure below.

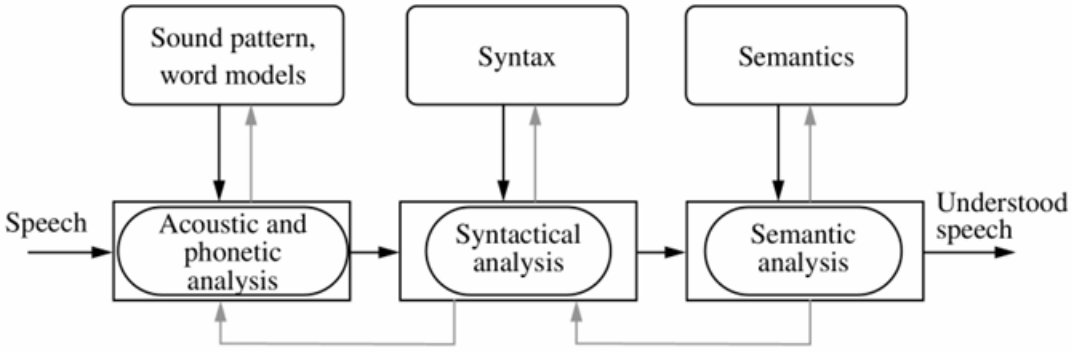


Figure: Speech recognition components

- In the first step, the principle is applied to a sound pattern and /or word model. An acoustical and phonetical analysis is performed.
- In the second step, certain speech units go through syntactical analysis. This process can detect errors of previous step and serves as an additional decision tool because the first step does not normally provide a final decision.
- The third step analyzes the semantics of the speech sequence recognized to this point. This step can detect errors from the previous decision process and remove them by using another interplay with other analytical methods. Note that even with current artificial intelligence and neural network technologies, the implementation of this step is extremely difficult.

Speech Transmission

- The area of speech transmission deals with efficient coding of the speech signal to allow speech/sound transmission at low transmission rates over networks, aiming to reduce perceptible quantity degradation.
- The goal is to provide the receiver with the same speech/sound quality as was generated at the sender side.
- This section includes some principles that are connected to speech generation and recognition.

1. **Signal Form Encoding/ Pulse Code Modulation (PCM)**
   - It does not consider speech-specific properties and parameters. The goal is to achieve the most efficient coding of the audio signal.
   - A straightforward technique for digitizing an analog signal (waveform) is PCM. This method is simple, but it still meets the high-quality demands stereo-audio signals in the data rate used for CDs:

$$\text{rate} = 2 \times \frac{44,100}{s} \times \frac{16\text{bits}}{8\text{bits/byte}} = 176,400 \text{ bytes/s} = 1,411,200 \text{ bytes/s}$$

2. **Source Coding**
   - Source encoding is a technique used in speech transmission to reduce the amount of data required for encoding signals by exploiting the inherent characteristics of the signal.
   - Parameterized systems work with source encoding algorithm. Here, the specific speech characteristics are used for data rate reduction.
   - Channel vo-coder is an example of such a parameterized system as shown in figure below.

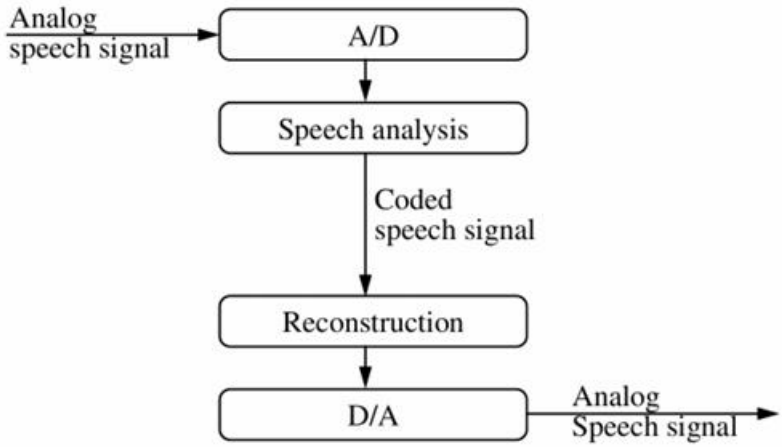


Figure: Components of a speech transmission system using source encoding

3. **Recognition/Synthesis Methods**
   - These methods are a cutting-edge approach in speech transmission aimed at significantly reducing data volumes while maintaining high quality.
   - There have been attempts to reduce the transmission rate using pure recognition/synthesis. Speech recognition follows on the sender side of a speech transmission system and speech synthesis (generation) follows on the receiver side as shown in figure below:

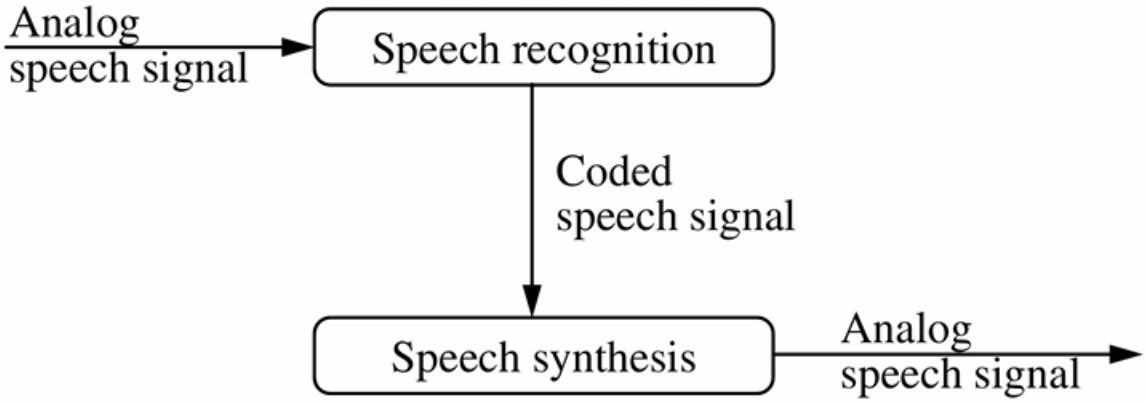


Figure: Components of a recognition-synthesis system for speech transmission

4. **Achieved Quality**
   - Achieved Quality in speech and audio transmission is a critical factor in multimedia systems, as it determines the minimum data rate required to maintain a defined level of audio fidelity.
   - Figure below demonstrates the relationship between audio quality and the number of bits per sampling value, showing how reducing the number of bits can still maintain good quality.

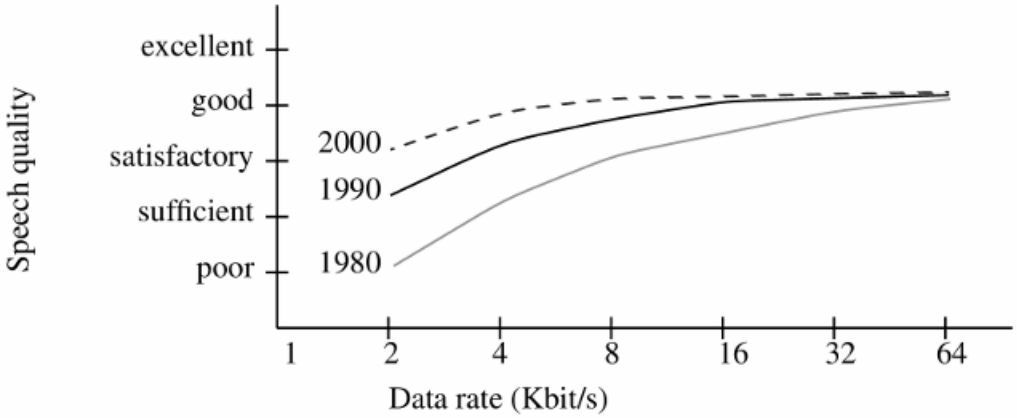


Figure: Dependence of the achieved speech quality on the data rate.